

Taylor and Maclaurin Series

ANSWER KEY



The Taylor series formula

- 1 State the general Taylor series formula for $f(x)$ centered at $x = a$.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots$$

- 2 Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x-1)^3$ term.

$$f(1) = 0; f'(x) = \frac{1}{x}, f'(1) = 1; f''(x) = -\frac{1}{x^2}, f''(1) = -1; f'''(x) = \frac{2}{x^3}, f'''(1) = 2. \text{ Series: } 0 + (x-1) - \frac{1}{2}(x-1)^2 + \frac{2}{6}(x-1)^3 = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}.$$

Common Maclaurin series

- 3 State the Maclaurin series for e^x , $\sin x$, and $\cos x$.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

- 4 Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2} .

$$\text{Substitute } x \rightarrow -x^2: e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} = 1 - x^2 + \frac{x^4}{2} - \dots$$

Using known series to build new series

- 5 Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x \sin x$.

$$x \sin x = x \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \right) = x^2 - \frac{x^4}{6} + \frac{x^6}{120} - \dots$$

- 6 Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.

$$\cos(0.1) \approx 1 - \frac{(0.1)^2}{2} + \frac{(0.1)^4}{24} = 1 - 0.005 + 0.0000042 \approx 0.995004.$$

The Lagrange error bound

- 7 State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n .

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}, \text{ where } M \text{ is an upper bound for } |f^{(n+1)}(z)| \text{ on the interval between } a \text{ and } x.$$

- 8 Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \leq 1$ for all z .

$$|R_3(0.5)| \leq \frac{1}{4!} |0.5|^4 = \frac{0.0625}{24} \approx 0.0026.$$

Radius of convergence of Taylor series

- 9 State the radius of convergence of the Maclaurin series for e^x , $\sin x$, and $\cos x$.

All three series have radius of convergence $R = \infty$ — they converge for all real x .

Visualizing Taylor polynomial approximations

- 10 Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.

The approximation is excellent near $x = 0$ (where the curves are nearly indistinguishable) and progressively worsens toward $x = \pm\pi$, since Taylor polynomials are most accurate near the center of expansion and lose accuracy farther away.

- 11 Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x .

Since e^x is convex (concave up) and its Taylor polynomial is tangent to it at $x = 0$ up to second order, $P_2(x)$ tends to underestimate e^x away from the center, with the gap growing faster on the positive side due to the exponential's rapid growth.

Building Taylor series at a nonzero center

- 12 Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.

$f(\frac{\pi}{2}) = 0$; $f'(x) = -\sin x$, $f'(\frac{\pi}{2}) = -1$; $f''(x) = -\cos x$, $f''(\frac{\pi}{2}) = 0$; $f'''(x) = \sin x$, $f'''(\frac{\pi}{2}) = 1$. Series:
 $-(x - \frac{\pi}{2}) + \frac{1}{6}(x - \frac{\pi}{2})^3 - \dots$

Applying the Lagrange Error Bound to bound accuracy requirements

- 13 Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \leq 1$, find the maximum error in approximating $\cos(0.3)$.

$|R_4(0.3)| \leq \frac{1}{5!}|0.3|^5 = \frac{0.00243}{120} \approx 0.0000203$ — an extremely small error, confirming the Maclaurin polynomial is highly accurate this close to the center.

The binomial series

- 14 State the binomial series for $(1+x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1+x}$ (i.e. $k = \frac{1}{2}$) up through the quadratic term.

$$(1+x)^k = 1 + kx + \frac{k(k-1)}{2}x^2 + \dots. \text{ With } k = \frac{1}{2}: \sqrt{1+x} = 1 + \frac{1}{2}x + \frac{(1/2)(-1/2)}{2}x^2 + \dots = 1 + \frac{x}{2} - \frac{x^2}{8} + \dots.$$

- 15 Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.

$$\sqrt{1.1} \approx 1 + \frac{0.1}{2} - \frac{(0.1)^2}{8} = 1 + 0.05 - 0.00125 = 1.04875 \text{ (true value } \approx 1.04881).$$

Deriving new Maclaurin series from known ones

- 16 Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots; e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \dots. \text{ Averaging cancels odd terms:}$$
$$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \dots.$$

Multiple choice — easy 15

- 17 Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^5 .

- a. $\frac{121}{40}$
- b. $\frac{81}{8}$
- c. $\frac{1}{40}$
- d. $\frac{81}{40}$ ✓

The coefficient of x^n is $\frac{3^n}{n!}$, so the coefficient of x^5 is $\frac{3^5}{5!} = \frac{243}{120} = \frac{81}{40}$.

- 18 Using $e^{7x} = \sum_{n=0}^{\infty} \frac{(7x)^n}{n!}$, find the coefficient of x^4 .

- a. $\frac{7}{24}$
- b. $\frac{2425}{24}$
- c. $\frac{2401}{6}$
- d. $\frac{2401}{24}$ ✓

The coefficient of x^n is $\frac{7^n}{n!}$, so the coefficient of x^4 is $\frac{7^4}{4!} = \frac{2401}{24} = \frac{2401}{24}$.

19 Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^2 .

a. 18 ✓

b. 19

c. 3

d. 36

The coefficient of x^n is $\frac{6^n}{n!}$, so the coefficient of x^2 is $\frac{6^2}{2!} = \frac{36}{2} = 18$.

20 Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{11}{2}$

b. $\frac{27}{2}$

c. $\frac{1}{2}$

d. $\frac{9}{2}$ ✓

The coefficient of x^n is $\frac{3^n}{n!}$, so the coefficient of x^3 is $\frac{3^3}{3!} = \frac{27}{6} = \frac{9}{2}$.

21 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^4 .

a. $\frac{512}{3}$ ✓

b. $\frac{2048}{3}$

c. $\frac{515}{3}$

d. $\frac{1}{3}$

The coefficient of x^n is $\frac{8^n}{n!}$, so the coefficient of x^4 is $\frac{8^4}{4!} = \frac{4096}{24} = \frac{512}{3}$.

22 Using $e^{4x} = \sum_{n=0}^{\infty} \frac{(4x)^n}{n!}$, find the coefficient of x^5 .

a. $\frac{143}{15}$

b. $\frac{128}{15}$ ✓

c. $\frac{1}{30}$

d. $\frac{128}{3}$

The coefficient of x^n is $\frac{4^n}{n!}$, so the coefficient of x^5 is $\frac{4^5}{5!} = \frac{1024}{120} = \frac{128}{15}$.

23 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{4}{3}$

b. 256

c. $\frac{259}{3}$

d. $\frac{256}{3}$ ✓

The coefficient of x^n is $\frac{8^n}{n!}$, so the coefficient of x^3 is $\frac{8^3}{3!} = \frac{512}{6} = \frac{256}{3}$.

24 Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^2 .

a. $\frac{11}{2}$

b. $\frac{9}{2}$ ✓

c. 9

d. $\frac{3}{2}$

The coefficient of x^n is $\frac{3^n}{n!}$, so the coefficient of x^2 is $\frac{3^2}{2!} = \frac{9}{2} = \frac{9}{2}$.

25 Using $e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$, find the coefficient of x^2 .

a. 4

b. 3

c. 2 ✓

d. 1

The coefficient of x^n is $\frac{2^n}{n!}$, so the coefficient of x^2 is $\frac{2^2}{2!} = \frac{4}{2} = 2$.

26 Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^3 .

a. 36 ✓

b. 108

c. 1

d. 37

The coefficient of x^n is $\frac{6^n}{n!}$, so the coefficient of x^3 is $\frac{6^3}{3!} = \frac{216}{6} = 36$.

27 Using $e^{4x} = \sum_{n=0}^{\infty} \frac{(4x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{2}{3}$

b. 32

c. $\frac{35}{3}$

d. $\frac{32}{3}$ ✓

The coefficient of x^n is $\frac{4^n}{n!}$, so the coefficient of x^3 is $\frac{4^3}{3!} = \frac{64}{6} = \frac{32}{3}$.

28 Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^5 .

a. 324

b. $\frac{324}{5}$ ✓

c. $\frac{329}{5}$

d. $\frac{1}{20}$

The coefficient of x^n is $\frac{6^n}{n!}$, so the coefficient of x^5 is $\frac{6^5}{5!} = \frac{7776}{120} = \frac{324}{5}$.

29 Using $e^{9x} = \sum_{n=0}^{\infty} \frac{(9x)^n}{n!}$, find the coefficient of x^3 .

a. $\frac{729}{2}$

b. $\frac{243}{2}$ ✓

c. $\frac{245}{2}$

d. $\frac{3}{2}$

The coefficient of x^n is $\frac{9^n}{n!}$, so the coefficient of x^3 is $\frac{9^3}{3!} = \frac{729}{6} = \frac{243}{2}$.

30 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^2 .

a. 32 ✓

b. 64

c. 33

d. 4

The coefficient of x^n is $\frac{8^n}{n!}$, so the coefficient of x^2 is $\frac{8^2}{2!} = \frac{64}{2} = 32$.

31 Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^5 .

a. $\frac{1}{15}$

b. $\frac{4096}{3}$

c. $\frac{4111}{15}$

d. $\frac{4096}{15}$ ✓

The coefficient of x^n is $\frac{8^n}{n!}$, so the coefficient of x^5 is $\frac{8^5}{5!} = \frac{32768}{120} = \frac{4096}{15}$.

Multiple choice — medium 15

32 Using $\sin(6x) = \sum_{n=0}^{\infty} \frac{(-1)^n (6x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. -35

b. 36

c. -36 ✓

d. 108

With $2n + 1 = 3$, $n = 1$: coefficient = $\frac{(-1)^1 6^3}{3!} = (-1) \cdot \frac{216}{6} = -36$.

33 Using $\sin(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

- a. $\frac{19}{15}$
- b. $\frac{4}{3}$
- c. $-\frac{4}{15}$
- d. $\frac{4}{15}$ ✓

With $2n + 1 = 5$, $n = 2$: coefficient $= \frac{(-1)^2 2^5}{5!} = (1) \cdot \frac{32}{120} = \frac{4}{15}$.

34 Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

- a. $\frac{125}{2}$
- b. $-\frac{125}{6}$ ✓
- c. $\frac{125}{6}$
- d. $-\frac{119}{6}$

With $2n + 1 = 3$, $n = 1$: coefficient $= \frac{(-1)^1 5^3}{3!} = (-1) \cdot \frac{125}{6} = -\frac{125}{6}$.

35 Using $\sin(9x) = \sum_{n=0}^{\infty} \frac{(-1)^n (9x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

- a. $\frac{19683}{8}$
- b. $-\frac{19683}{40}$
- c. $\frac{19723}{40}$
- d. $\frac{19683}{40}$ ✓

With $2n + 1 = 5$, $n = 2$: coefficient $= \frac{(-1)^2 9^5}{5!} = (1) \cdot \frac{59049}{120} = \frac{19683}{40}$.

36 Using $\sin(4x) = \sum_{n=0}^{\infty} \frac{(-1)^n (4x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

- a. $-\frac{29}{3}$
- b. $\frac{32}{3}$
- c. 32
- d. $-\frac{32}{3}$ ✓

With $2n + 1 = 3$, $n = 1$: coefficient $= \frac{(-1)^1 4^3}{3!} = (-1) \cdot \frac{64}{6} = -\frac{32}{3}$.

37 Using $\sin(7x) = \sum_{n=0}^{\infty} \frac{(-1)^n (7x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $-\frac{337}{6}$

b. $-\frac{343}{6}$ ✓

c. $\frac{343}{6}$

d. $\frac{343}{2}$

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 7^3}{3!} = (-1) \cdot \frac{343}{6} = -\frac{343}{6}$.

38 Using $\sin(4x) = \sum_{n=0}^{\infty} \frac{(-1)^n (4x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $\frac{128}{15}$ ✓

b. $\frac{143}{15}$

c. $-\frac{128}{15}$

d. $\frac{128}{3}$

With $2n+1=5$, $n=2$: coefficient $= \frac{(-1)^2 4^5}{5!} = (1) \cdot \frac{1024}{120} = \frac{128}{15}$.

39 Using $\sin(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. 4

b. $-\frac{1}{3}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$ ✓

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 2^3}{3!} = (-1) \cdot \frac{8}{6} = -\frac{4}{3}$.

40 Using $\sin(6x) = \sum_{n=0}^{\infty} \frac{(-1)^n (6x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $\frac{324}{5}$ ✓

b. $-\frac{324}{5}$

c. $\frac{329}{5}$

d. 324

With $2n+1=5$, $n=2$: coefficient $= \frac{(-1)^2 6^5}{5!} = (1) \cdot \frac{7776}{120} = \frac{324}{5}$.

41 Using $\sin(9x) = \sum_{n=0}^{\infty} \frac{(-1)^n (9x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $-\frac{241}{2}$

b. $-\frac{243}{2}$ ✓

c. $\frac{729}{2}$

d. $\frac{243}{2}$

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 9^3}{3!} = (-1) \cdot \frac{729}{6} = -\frac{243}{2}$.

42 Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

a. $-\frac{81}{40}$

b. $\frac{81}{8}$

c. $\frac{121}{40}$

d. $\frac{81}{40}$ ✓

With $2n+1=5$, $n=2$: coefficient $= \frac{(-1)^2 3^5}{5!} = (1) \cdot \frac{243}{120} = \frac{81}{40}$.

43 Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

a. $\frac{9}{2}$

b. $\frac{27}{2}$

c. $-\frac{7}{2}$

d. $-\frac{9}{2}$ ✓

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 3^3}{3!} = (-1) \cdot \frac{27}{6} = -\frac{9}{2}$.

44 Using $\sin(8x) = \sum_{n=0}^{\infty} \frac{(-1)^n (8x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3 .

- a. $\frac{256}{3}$
- b. $-\frac{253}{3}$
- c. $-\frac{256}{3}$ ✓
- d. 256

With $2n + 1 = 3$, $n = 1$: coefficient $= \frac{(-1)^1 8^3}{3!} = (-1) \cdot \frac{512}{6} = -\frac{256}{3}$.

45 Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

- a. $-\frac{625}{24}$
- b. $\frac{649}{24}$
- c. $\frac{625}{24}$ ✓
- d. $\frac{3125}{24}$

With $2n + 1 = 5$, $n = 2$: coefficient $= \frac{(-1)^2 5^5}{5!} = (1) \cdot \frac{3125}{120} = \frac{625}{24}$.

46 Using $\sin(8x) = \sum_{n=0}^{\infty} \frac{(-1)^n (8x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5 .

- a. $-\frac{4096}{15}$
- b. $\frac{4096}{3}$
- c. $\frac{4096}{15}$ ✓
- d. $\frac{4111}{15}$

With $2n + 1 = 5$, $n = 2$: coefficient $= \frac{(-1)^2 8^5}{5!} = (1) \cdot \frac{32768}{120} = \frac{4096}{15}$.

Multiple choice — hard 15

47 Use the Maclaurin series for e^u (with $u = 3x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{3x^2} .

- a. $\frac{2187}{8}$
- b. $\frac{9}{4480}$
- c. $\frac{35}{8}$
- d. $\frac{27}{8}$ ✓

Substituting $u = 3x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^8 term comes from $n = 4$: $\frac{(3x^2)^4}{4!} = \frac{81}{24}x^8$. Coefficient: $\frac{27}{8}$.

48 Use the Maclaurin series for e^u (with $u = 4x^2$) to find the coefficient of x^6 in the Maclaurin series for e^{4x^2} .

- a. $\frac{32}{3}$ ✓
- b. $\frac{2048}{3}$
- c. $\frac{4}{45}$
- d. $\frac{35}{3}$

Substituting $u = 4x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^6 term comes from $n = 3$: $\frac{(4x^2)^3}{3!} = \frac{64}{6}x^6$. Coefficient: $\frac{32}{3}$.

49 Use the Maclaurin series for e^u (with $u = 2x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{2x^2} .

- a. $\frac{2}{3}$ ✓
- b. $\frac{1}{2520}$
- c. $\frac{5}{3}$
- d. $\frac{32}{3}$

Substituting $u = 2x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^8 term comes from $n = 4$: $\frac{(2x^2)^4}{4!} = \frac{16}{24}x^8$. Coefficient: $\frac{2}{3}$.

50 Use the Maclaurin series for e^u (with $u = 9x^2$) to find the coefficient of x^6 in the Maclaurin series for e^{9x^2} .

- a. $\frac{245}{2}$
- b. $\frac{81}{80}$
- c. $\frac{243}{2}$ ✓
- d. $\frac{177147}{2}$

Substituting $u = 9x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^6 term comes from $n = 3$: $\frac{(9x^2)^3}{3!} = \frac{729}{6}x^6$. Coefficient: $\frac{243}{2}$.

51 Use the Maclaurin series for e^u (with $u = 7x^2$) to find the coefficient of x^4 in the Maclaurin series for e^{7x^2} .

a. $\frac{49}{24}$

b. $\frac{2401}{2}$

c. $\frac{49}{2}$ ✓

d. $\frac{51}{2}$

Substituting $u = 7x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^4 term comes from $n = 2$: $\frac{(7x^2)^2}{2!} = \frac{49}{2}x^4$. Coefficient: $\frac{49}{2}$.

52 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 6$, $n = 2$, and $|x-a| = 1$, find the error bound.

a. 2

b. $\frac{7}{6}$

c. 3

d. 1 ✓

$$|R_2(x)| \leq \frac{6}{(2+1)!}(1)^{2+1} = \frac{6}{6} = 1.$$

53 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 9$, $n = 2$, and $|x-a| = 1$, find the error bound.

a. $\frac{5}{3}$

b. $\frac{3}{2}$ ✓

c. $\frac{5}{2}$

d. $\frac{9}{2}$

$$|R_2(x)| \leq \frac{9}{(2+1)!}(1)^{2+1} = \frac{9}{6} = \frac{3}{2}.$$

- 54 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1}$ with $M = 5$, $n = 3$, and $|x - a| = 1$, find the error bound.

a. $\frac{5}{6}$

b. $\frac{5}{24}$ ✓

c. $\frac{29}{24}$

d. $\frac{1}{4}$

$$|R_3(x)| \leq \frac{5}{(3+1)!} (1)^{3+1} = \frac{5}{24} = \frac{5}{24}.$$

- 55 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1}$ with $M = 5$, $n = 4$, and $|x - a| = 1$, find the error bound.

a. $\frac{1}{24}$ ✓

b. $\frac{5}{24}$

c. $\frac{1}{20}$

d. $\frac{25}{24}$

$$|R_4(x)| \leq \frac{5}{(4+1)!} (1)^{4+1} = \frac{5}{120} = \frac{1}{24}.$$

- 56 Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1}$ with $M = 4$, $n = 2$, and $|x - a| = 1$, find the error bound.

a. $\frac{5}{3}$

b. 2

c. $\frac{5}{6}$

d. $\frac{2}{3}$ ✓

$$|R_2(x)| \leq \frac{4}{(2+1)!} (1)^{2+1} = \frac{4}{6} = \frac{2}{3}.$$

57 Using $\cosh(2x) = \frac{e^{2x} + e^{-2x}}{2} = \sum_{n=0}^{\infty} \frac{(2x)^{2n}}{(2n)!}$, find the coefficient of x^2 .

a. 4

b. $\frac{2}{3}$

c. 2 ✓

d. 3

With $2n = 2$: coefficient $= \frac{2^2}{2!} = \frac{4}{2} = 2$.

58 Using $\cosh(8x) = \frac{e^{8x} + e^{-8x}}{2} = \sum_{n=0}^{\infty} \frac{(8x)^{2n}}{(2n)!}$, find the coefficient of x^6 .

a. $\frac{32768}{15}$

b. $\frac{16384}{315}$

c. $\frac{16384}{45}$ ✓

d. $\frac{16429}{45}$

With $2n = 6$: coefficient $= \frac{8^6}{6!} = \frac{262144}{720} = \frac{16384}{45}$.

59 Using $\cosh(6x) = \frac{e^{6x} + e^{-6x}}{2} = \sum_{n=0}^{\infty} \frac{(6x)^{2n}}{(2n)!}$, find the coefficient of x^2 .

a. 18 ✓

b. 36

c. 6

d. 19

With $2n = 2$: coefficient $= \frac{6^2}{2!} = \frac{36}{2} = 18$.

60 Using $\cosh(4x) = \frac{e^{4x} + e^{-4x}}{2} = \sum_{n=0}^{\infty} \frac{(4x)^{2n}}{(2n)!}$, find the coefficient of x^4 .

a. $\frac{35}{3}$

b. $\frac{128}{3}$

c. $\frac{32}{3}$ ✓

d. $\frac{32}{15}$

With $2n = 4$: coefficient $= \frac{4^4}{4!} = \frac{256}{24} = \frac{32}{3}$.

61 Using $\cosh(7x) = \frac{e^{7x} + e^{-7x}}{2} = \sum_{n=0}^{\infty} \frac{(7x)^{2n}}{(2n)!}$, find the coefficient of x^6 .

a. $\frac{118369}{720}$

b. $\frac{117649}{120}$

c. $\frac{16807}{720}$

d. $\frac{117649}{720}$ ✓

With $2n = 6$: coefficient $= \frac{7^6}{6!} = \frac{117649}{720} = \frac{117649}{720}$.